3.2.3 Classical Gram-Schmidt algorithm

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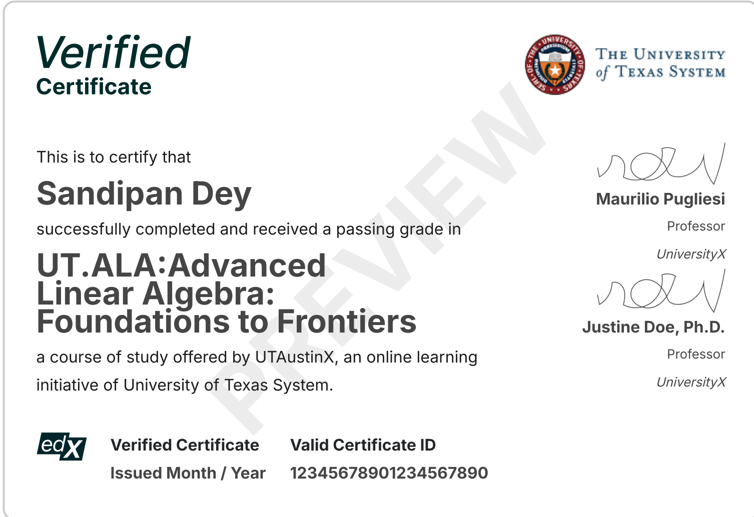

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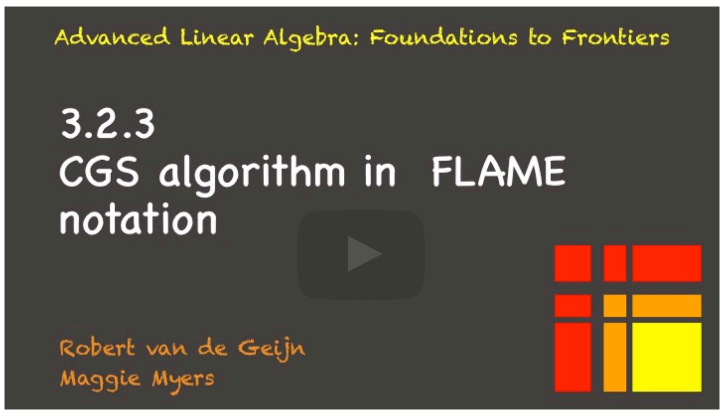

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2.0x

The way we've described the Gram Schmidt Orthogonalization process, and of course

since that computes the QR factorization, therefore how to compute the QR

factorization, involves a lot of indices. And our experience is that

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Reading assignment

1.0/1.0 point (graded)

Read Unit 3.2.3 of the notes. [LINK]

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? Possible administrative error in HW Assignment--wrong MATLAB file
Instructions call for implementing the function CGS_QR(A) in the MATLAB fil... 2

Q Transpose
Hey! I have a question: So, the below algorithm used Q Transpose instead o... 3

Homework 3.2.3.1

1.0/1.0 point (graded)
Implement the algorithm

$[Q,R] = \text{CGS\_QR}(A)$

$A \rightarrow \left(A_L \mid A_R \right), Q \rightarrow \left(Q_L \mid Q_R \right), R \rightarrow \left(\begin{array}{c|c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right)$

A_L and Q_L has 0 columns and R_{TL} is 0×0

while $n(A_L) < n(A)$

$\left(A_L \mid A_R \right) \rightarrow \left(A_0 \mid a_1 \quad A_2 \right), \left(Q_L \mid Q_R \right) \rightarrow \left(Q_0 \mid q_1 \quad Q_2 \right),$

$\left(\begin{array}{c|c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|cc} R_{00} & r_{01} & R_{02} \\ \hline 0 & \rho_{11} & r_{12}^T \\ 0 & 0 & R_{22} \end{array} \right)$

$r_{01} := Q_0^T a_1$

$a_1^\perp := a_1 - Q_0 r_{01}$

$\rho_{11} := \|a_1^\perp\|_2$

$q_1 := a_1^\perp / \rho_{11}$

$\left(A_L \mid A_R \right) \leftarrow \left(A_0 \quad a_1 \mid A_2 \right), \left(Q_L \mid Q_R \right) \leftarrow \left(Q_0 \quad q_1 \mid Q_2 \right),$

$\left(\begin{array}{c|c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right) \leftarrow \left(\begin{array}{cc|c} R_{00} & r_{01} & R_{02} \\ 0 & \rho_{11} & r_{12}^T \\ \hline 0 & 0 & R_{22} \end{array} \right)$

endwhile

as

```
function [ Q, R ] = CGS_QR( A )
```

by completing the code in Assignments/Week03/matlab/CGS_QR.m. Input is an $m \times n$ matrix A. Output is the matrix Q and the upper triangular matrix R. You may want to use Assignments/Week03/matlab/test_CGS_QR.m to check your implementation.

- [x] done

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